

Basic Gates:

basic rules; 1 represents light up (on).

0 represents light down (off).

*In red stone, you could see it directly

1) Not gate: The not gate is when the output opposite to the input.

For example: If you input 1, the output will be 0.

If you input 0, the output will be 1.

Note: standard not gate only have 1 input and it's called "inverter."

Diagram: A



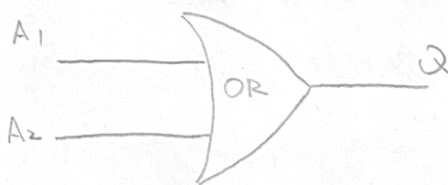
Input → A	Q → Output
0	1
1	0

2) Or gate: The or gate it's interesting, the output will be 0 if and only if all input is 0, if there's a input it's 1, the output will be 1.

For example: Imagine you have two inputs. If one of them is 1, you will have 1 for the output, if you have both of them is 1, you will have 1 as well, but if your inputs are neither of them are 1, you will have 0 at the end.

Note: Standard OR gate could have more than one input, so you could have three or four input in this gate.

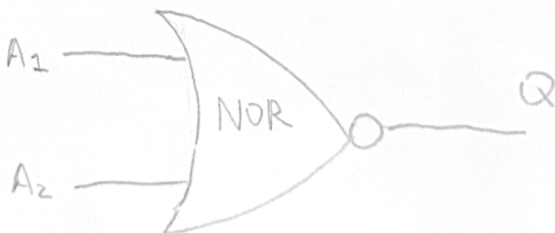
Diagram:



A ₁	A ₂	Q
0	0	0
1	0	1
0	1	1
1	1	1

3) Nor Gate: This one is simple, it's the reverse of or gate, so the output will be opposite with the same input you plugged in for or gate.

Diagram:



A ₁	A ₂	Q
0	0	1
1	0	0
0	1	0
1	1	0

4) And Gate: The behavior of And gate might be easy to understand based off its name "AND". You must put both inputs as 1 in order to have the 1 for the output, otherwise, it's a 0.

Note: Same as or gate, you could have more than one input as well

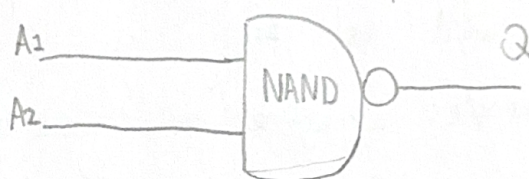
Diagram:



A ₁	A ₂	Q
0	0	0
1	0	0
0	1	0
1	1	1

5) NAND Gate: Based on its name, it means that Not And Gate, which is the reverse of And gate. It's like or gate but when inputs are all 0, it will give 1 for the output

Diagram:

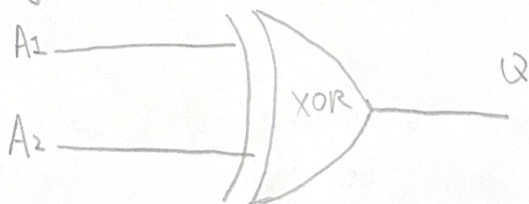


A ₁	A ₂	Q
0	0	1
1	0	1
0	1	1
1	1	0

6) XOR Gate: XOR means that exclusive or gate, it will give 1 as output if you have different input.

For example: Imagine you have three inputs (I want to make some difference with two outputs) If you have 0,0,0, you will have 0 at the end, and 1,1,1, still at the end output is 0, but any combination besides that are 1 for the output.

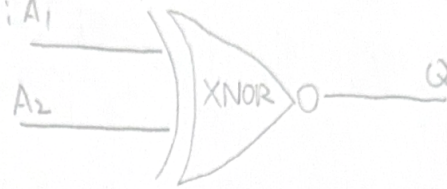
Diagram (two inputs)



A ₁	A ₂	Q
0	0	0
1	0	1
0	1	1
1	1	0

Xnor Gate (ExclusiveNor, gate): The exclusive nor gates are if inputs are different, it's 0, otherwise it's 1,

Diagram: A_1



A_1	A_2	Q
0	0	1
1	0	0
0	1	0
1	1	1